

Marine N-3 Polyunsaturated Fatty Acids Are Inversely Associated with Risk of Type 2 Diabetes in Asians: A Systematic Review and Meta-Analysis

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Background Prospective cohort studies in relation to the associations between n-3 polyunsaturated fatty acids (PUFA) and risk of type 2 diabetes (T2D) were inconsistent. Differences in tissue n-3 PUFA compositions in subjects with and without T2D were also inconsistent in both cohort and case-control studies. We conducted a systematic review and meta-analysis of prospective cohort studies to examine the associations of fish and n-3 PUFA intake with T2D risk. The differences in tissue n-3 PUFA compositions in subjects with and without T2D were investigated based on cohort and case-control studies. Methods and Findings PubMed, Embase, Cochrane library, China National Knowledge Infrastructure (CNKI) and Chinese VIP database up to January 2012 was used to identify relevant studies, and reference lists from retrieved studies were reviewed. Two authors independently extracted the data. Random-effects models were used to pool the summary relative risk (RR). Twenty-four studies including 24,509 T2D patients and 545,275 participants were identified. For cohort studies, the summary RR of T2D for the highest vs lowest categories of total fish, marine n-3 PUFA and alpha-linolenic acid intake was 1.07 (95% CI: 0.91, 1.25), 1.07 (95% CI: 0.95, 1.20) and 0.93 (95% CI: 0.81, 1.07), respectively. Subgroup analyses indicated that summary RR (highest vs lowest category) of T2D for fish and marine n-3 PUFA intake was 0.89 (95% CI: 0.81, 0.98) and 0.87 (95% CI: 0.79, 0.96) for Asian populations, and 1.20 (95% CI: 1.01, 1.44) and 1.16 (95% CI: 1.04, 1.28) for Western populations. Asian subjects with T2D had significantly lower tissue compositions of C22 Δ 6n-3 (SMD: -1.43; 95% CI: -1.75, -1.12) and total n-3 PUFA (SMD: -1.41; 95% CI: -2.23, -0.59) compared with those without T2D. Conclusion This systematic review and meta-analysis provides evidence that marine n-3 PUFA have beneficial effects on the prevention of T2D in Asian populations.